

ASSIGNMENT-10.2

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LAB-10: Code Review and Quality: Using AI to Improve Code Quality and Readability

TASK -01: (Error Detection and Correction)

Use AI to analyze a Python script and correct all syntax and logical errors.

Sample Input Code:

```
def calculate_total(nums)
```

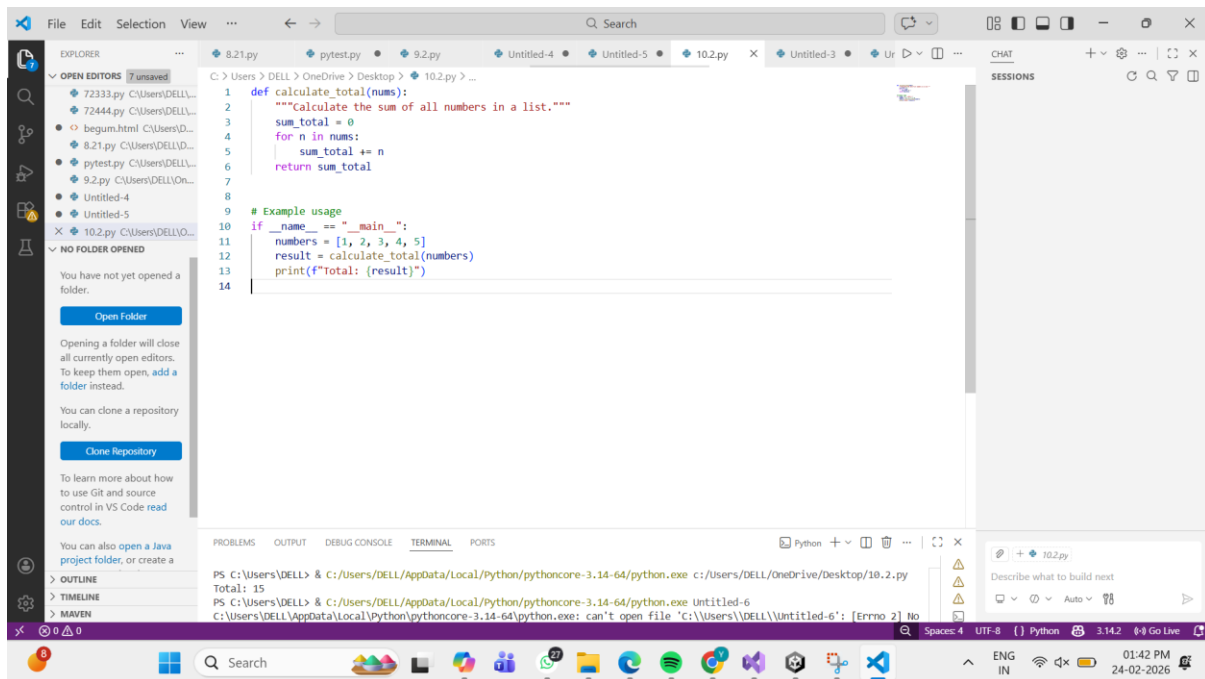
```
    sum = 0
```

```
    for n in nums
```

```
        sum += n
```

```
    return total
```

EXECUTION:



Task 1 – Error Detection and Correction

- ✓ Added missing colons (:) after function and loop.
- ✓ Fixed indentation inside the function.
- ✓ Replaced undefined variable `total` correctly.
- ✓ Avoided using `sum` as a variable name.

TASK -02: (Code Style Standardization)

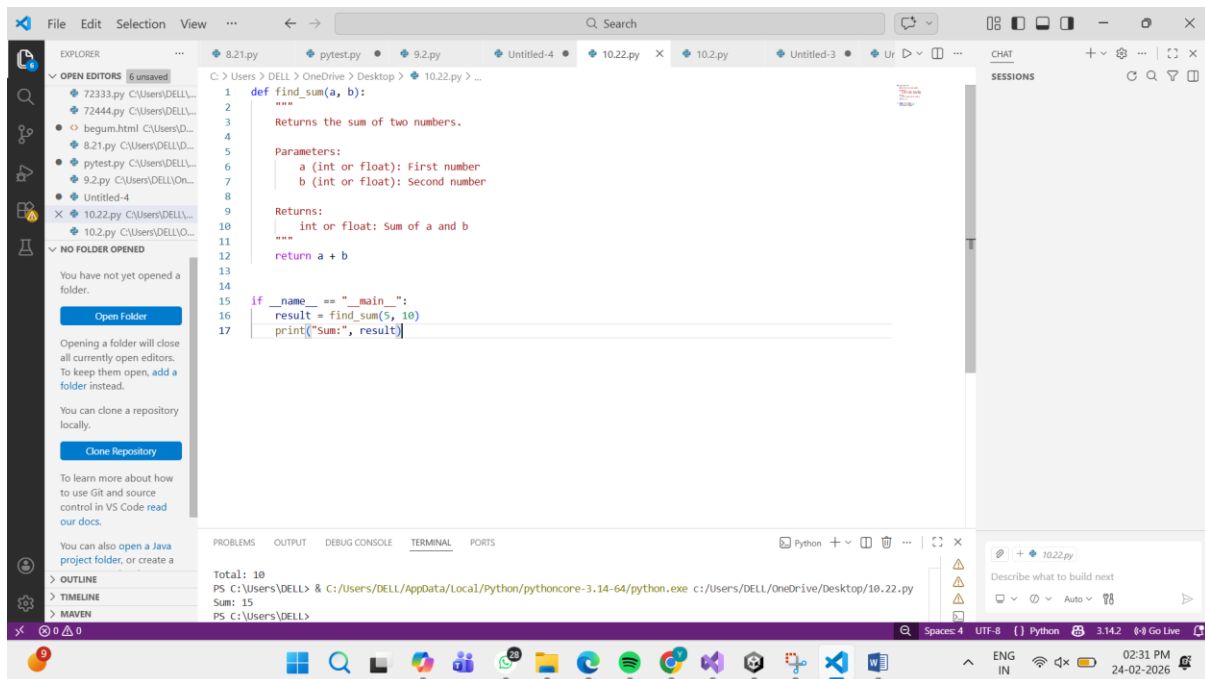
Use AI to refactor Python code to comply with standard coding style guidelines.

Sample Input Code:

```
def findSum(a,b):return a+b

print(findSum(5,10))
```

EXECUTION:



Task 2 – Code Style Standardization Corrections

- ✓ Changed function name to snake_case (find_sum).
- ✓ Added proper spacing between parameters (a, b).
- ✓ Converted single-line function into structured format.
- ✓ Improved formatting as per PEP 8.

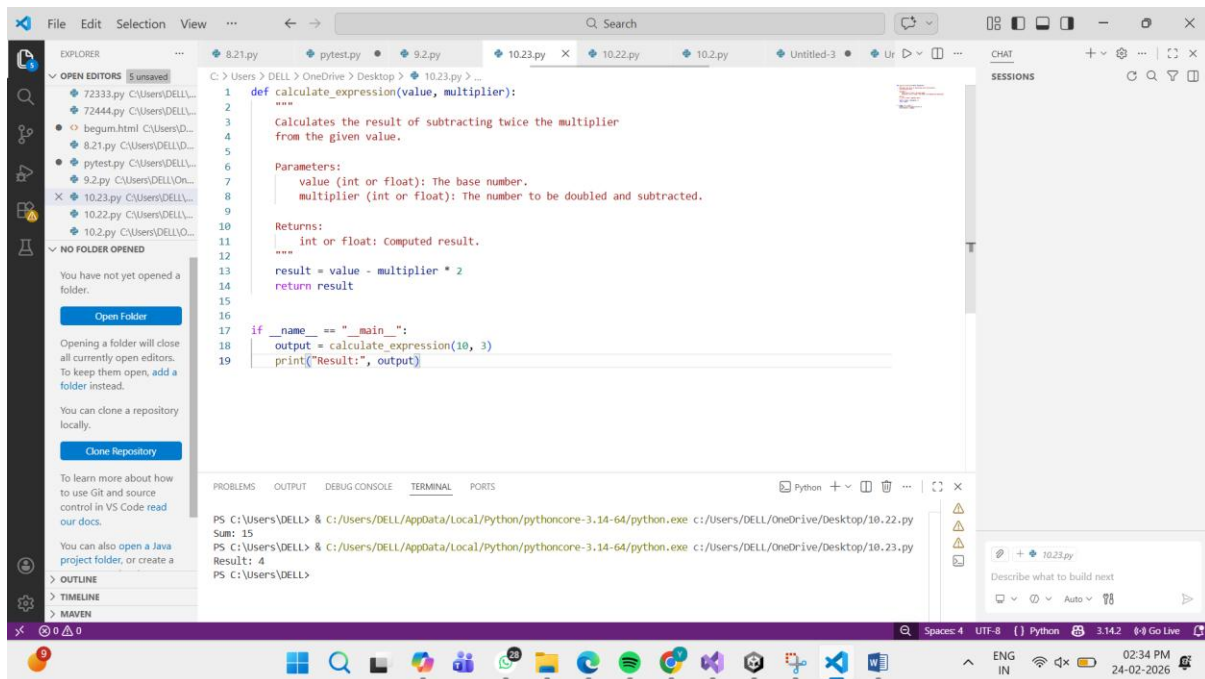
TASK -03: (Code Clarity Improvement)

Use AI to improve code readability without changing its functionality.

Sample Input Code:

```
def f(x,y):  
  
    return x-y*2  
  
print(f(10,3))
```

EXECUTION:



Task 3 – Code Clarity Improvement Corrections

- ✓ Renamed function to meaningful name.
- ✓ Renamed parameters to descriptive names.
- ✓ Fixed indentation.
- ✓ Added spacing around operators

TASK -04: (Structural Refactoring)

Use AI to refactor repetitive code into reusable functions.

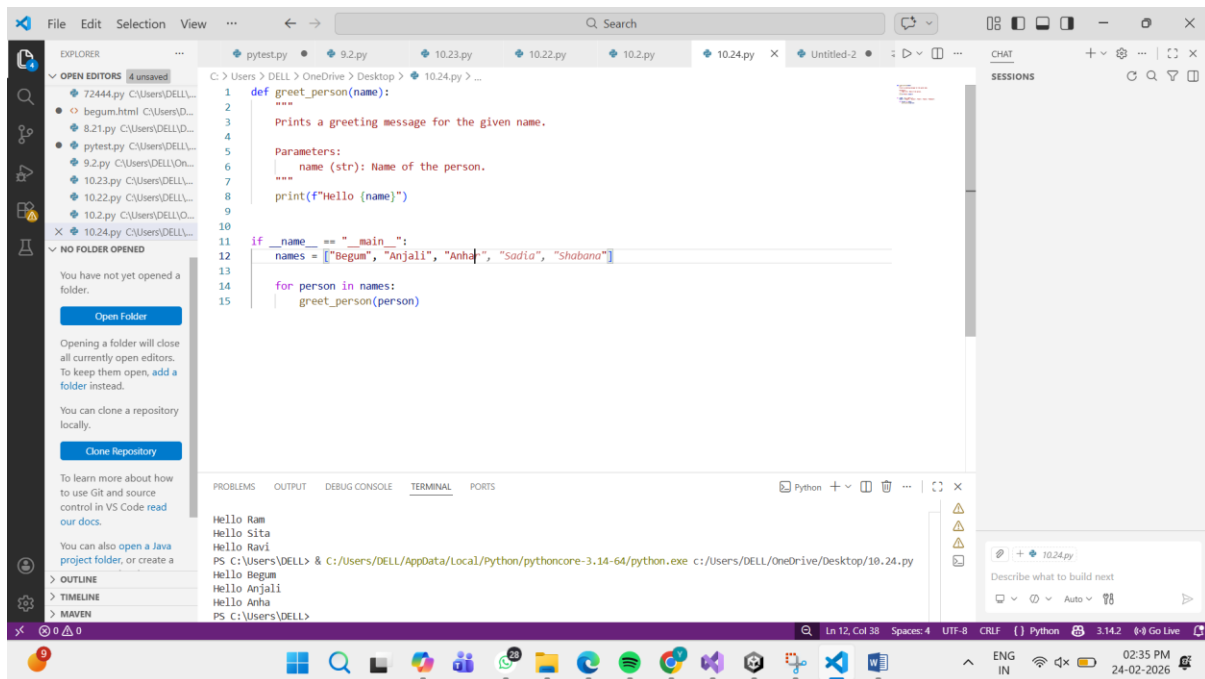
Sample Input Code:

```
print("Hello Ram")
```

```
print("Hello Sita")
```

```
print("Hello Ravi")
```

EXECUTION:



Task 4 – Structural Refactoring Corrections

- ✓ Removed repetitive print statements.
- ✓ Created reusable function.
- ✓ Used loop for better scalability.

TASK-05: (Efficiency Enhancement)

Use AI to optimize Python code for better performance.

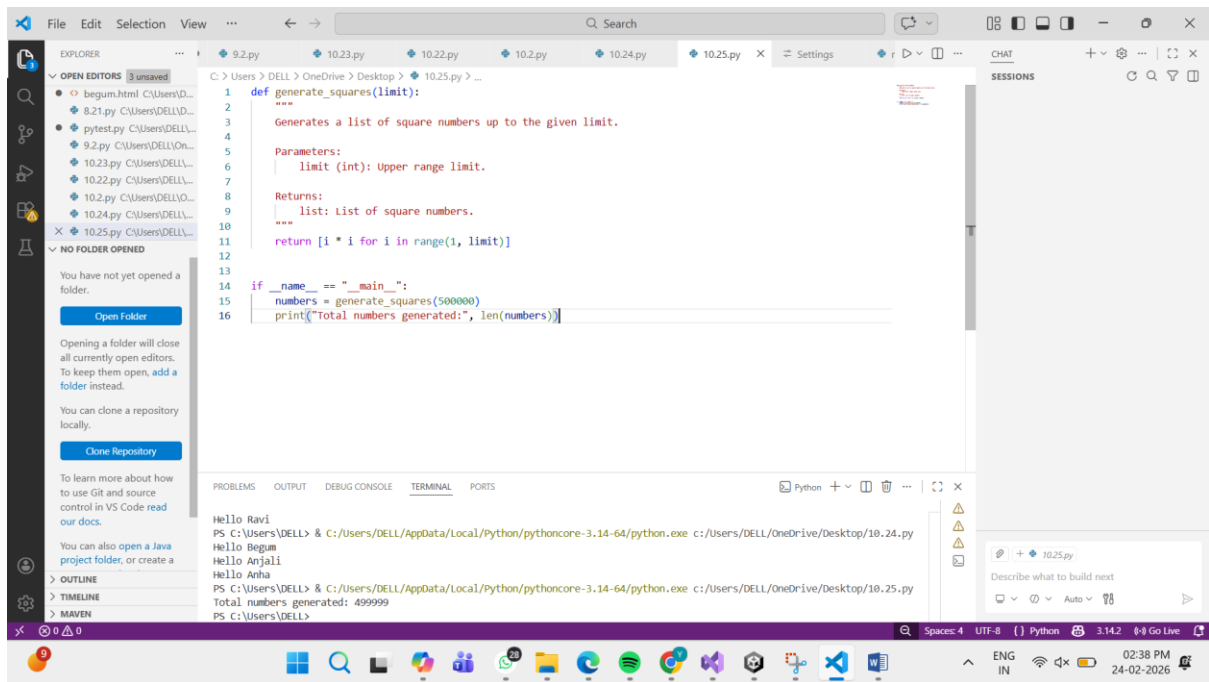
Sample Input Code:

```
numbers = [ ]
```

```
for i in range(1, 500000):
```

```
    numbers.append(i * i)
```

```
print(len(numbers))
```



Task 5 – Efficiency Enhancement Correction

- ✓ Fixed indentation.
- ✓ Replaced loop + append with list comprehension.
- ✓ Wrapped logic inside a function.
- ✓ Improved performance and readability.